

ORF 524 — Final Exam

Matias D. Cattaneo
Department of Operations Research and Financial Engineering
Princeton University

December 12, 2022

Contents

1 Semiparametric Bandwidth Selection (50 Points)	1
2 Optimal Conditional Z-estimation (50 Points)	7

Notes:

- Duration of examination: 180 minutes.
An additional 15 minutes are given to upload your solutions to Gradescope.
- All parts are equally weighted (5 points).
Please provide as much detail as possible in your answers.
Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
A single PDF file must be submitted to Gradescope.
- Only one page (double-sided) of personal notes are allowed during the examination.
This is otherwise a closed-book and individual examination.

Question 1: Semiparametric Bandwidth Selection (50 Points)

This question discusses bandwidth selection for kernel-based semiparametric and nonparametric estimators. Let x_i for $1 \leq i \leq n$ be i.i.d. real-valued random variables with Lebesgue density $f(x)$. Consider estimating the average density $\theta = \mathbb{E}[f(x_i)]$ using

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n \hat{f}_{i,n}(x_i), \quad \hat{f}_{i,n}(x) = \frac{1}{n-1} \sum_{j=1, j \neq i}^n \frac{1}{h} K\left(\frac{x_j - x}{h}\right),$$

where $K : \mathbb{R} \rightarrow \mathbb{R}$ is an even, bounded and compactly supported kernel function with $\int_{\mathbb{R}} K(u) du = 1$, and h is a positive bandwidth sequence with $h \rightarrow 0$ as $n \rightarrow \infty$. The goal of this question is to derive a data-driven procedure for selecting the bandwidth h . We take a plug-in approach based on mean squared error. Recall that the Hoeffding decomposition gives $\hat{\theta}_n = \theta_n + \bar{L}_n + \bar{Q}_n$ where

$$\begin{aligned} \theta_n &= \mathbb{E}[U_n(x_i, x_j)], & U_n(x_i, x_j) &= \frac{1}{h} K\left(\frac{x_i - x_j}{h}\right), \\ \bar{L}_n &= \frac{1}{n} \sum_{i=1}^n L_n(x_i), & L_n(x_i) &= 2(\mathbb{E}[U_n(x_i, x_j) \mid x_i] - \theta_n), \\ \bar{Q}_n &= \binom{n}{2}^{-1} \sum_{i=1}^{n-1} \sum_{j=i+1}^n Q_n(x_i, x_j), & Q_n(x_i, x_j) &= U_n(x_i, x_j) - \frac{1}{2}L_n(x_i) - \frac{1}{2}L_n(x_j) - \theta_n. \end{aligned}$$

1. Show that $\mathbb{E}[\bar{L}_n] = 0$, $\mathbb{E}[\bar{Q}_n] = 0$ and $\mathbb{E}[\bar{L}_n \bar{Q}_n] = 0$.

Solution: For $\bar{L}_n$,

$$\mathbb{E}[\bar{L}_n] = \mathbb{E}[L_n(x_i)] = 2\mathbb{E}[\mathbb{E}[U_n(x_i, x_j) \mid x_i] - \theta_n] = 2\mathbb{E}[U_n(x_i, x_j)] - 2\theta_n = 2\theta_n - 2\theta_n = 0$$

by the law of iterated expectations. Similarly for $\bar{Q}_n$,

$$\begin{aligned} \mathbb{E}[\bar{Q}_n] &= \mathbb{E}[Q_n(x_i, x_j)] = \mathbb{E}[U_n(x_i, x_j) - \mathbb{E}[U_n(x_i, x_j) \mid x_i] - \mathbb{E}[U_n(x_i, x_j) \mid x_j] + \theta_n] \\ &= \mathbb{E}[U_n(x_i, x_j)] - \mathbb{E}[U_n(x_i, x_j)] - \mathbb{E}[U_n(x_i, x_j)] + \theta_n = \theta_n - \theta_n - \theta_n + \theta_n = 0. \end{aligned}$$

Note that by independence we have $\mathbb{E}[Q_n(x_i, x_j)L_n(x_r)] = 0$ for i, j , and r distinct, and also

$$\begin{aligned} \mathbb{E}[Q_n(x_i, x_j)L_n(x_i)] &= \mathbb{E}[L_n(x_i)\mathbb{E}[Q_n(x_i, x_j) \mid x_i]] \\ &= \mathbb{E}\left[L_n(x_i)\mathbb{E}\left[U_n(x_i, x_j) - \frac{1}{2}L_n(x_i) - \frac{1}{2}L_n(x_j) - \theta_n \mid x_i\right]\right] \\ &= \mathbb{E}\left[L_n(x_i)\mathbb{E}\left[U_n(x_i, x_j) - \mathbb{E}[U_n(x_i, x_j) \mid x_i] + \theta_n - 0 - \theta_n \mid x_i\right]\right] = 0. \end{aligned}$$

Hence $\mathbb{E}[\bar{L}_n \bar{Q}_n] = 0$.

2. Show that $\mathbb{E}[U_n(x_i, x_j) \mid x_i] = \int_{\mathbb{R}} K(u)f(x_i + uh) du$ almost surely.

Solution: The i.i.d. assumption and a change of variables with $u = \frac{x_j - x_i}{h}$ gives

$$\begin{aligned}\mathbb{E}[U_n(x_i, x_j) \mid x_i] &= \int_{\mathbb{R}} \frac{1}{h} K\left(\frac{x_i - x_j}{h}\right) f(x_j) dx_j = \int_{\mathbb{R}} K(-u) f(x_i + uh) du \\ &= \int_{\mathbb{R}} K(u) f(x_i + uh) du\end{aligned}$$

by symmetry of the kernel.

3. Suppose the kernel K is of order $P \geq 2$, that is, $\int_{\mathbb{R}} u^p K(u) du = 0$ for all $1 \leq p \leq P-1$. Provide conditions on f such that $\mathbb{E}[\hat{\theta}_n] = \theta + h^P \mathcal{B} + o(h^P)$ and identify the bias $\mathcal{B}$.

Solution:

Assume f is $P+1$ times continuously differentiable with $f^{(r)}$ bounded for all $0 \leq r \leq P+1$. Then Taylor's theorem up to order P with Lagrange remainder gives

$$\begin{aligned}\mathbb{E}[\hat{\theta}_n] &= \mathbb{E}[\mathbb{E}[U_n(x_i, x_j) \mid x_i]] \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} K(u) f(x_i + uh) du f(x_i) dx_i \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} K(u) \left(\sum_{p=0}^P \frac{h^p u^p}{p!} f^{(p)}(x_i) + \frac{h^{P+1} u^{P+1}}{(P+1)!} f^{(P+1)}(\tilde{x}_i) \right) du f(x_i) dx_i \\ &= \int_{\mathbb{R}} f(x_i)^2 dx_i + \frac{h^P}{P!} \int_{\mathbb{R}} u^P K(u) du \int_{\mathbb{R}} f^{(P)}(x_i) f(x_i) dx_i + O(h^{P+1}) \\ &= \theta + \frac{h^P}{P!} \left(\int_{\mathbb{R}} u^P K(u) du \right) \mathbb{E}[f^{(P)}(x_i)] + o(h^P),\end{aligned}$$

where $\tilde{x}_i$ is between x_i and $x_i + uh$. Thus the leading bias is

$$\mathcal{B} = \frac{1}{P!} \left(\int_{\mathbb{R}} u^P K(u) du \right) \mathbb{E}[f^{(P)}(x_i)].$$

4. Show that $\mathbb{V}[\bar{L}_n] = \frac{\sigma^2}{n} + O\left(\frac{h^P}{n}\right)$ and give the value of σ^2 .

Solution: By the i.i.d. property,

$$\begin{aligned}\mathbb{V}[\bar{L}_n] &= \frac{1}{n} \mathbb{E}[L_n(x_i)^2] = \frac{4}{n} \mathbb{E}[(\mathbb{E}[U_n(x_i, x_j) \mid x_i] - \theta_n)^2] \\ &= \frac{4}{n} \mathbb{E}[\mathbb{E}[U_n(x_i, x_j) \mid x_i]^2] - \frac{4}{n} \theta_n^2.\end{aligned}$$

By part 2,

$$\mathbb{E}[(\mathbb{E}[U_n(x_i, x_j) \mid x_i])^2] = \mathbb{E} \left[\left(\int_{\mathbb{R}} K(u) f(x_i + uh) du \right)^2 \right].$$

Again by Taylor's theorem up to order $P - 1$ with Lagrange remainder,

$$\begin{aligned}\int_{\mathbb{R}} K(u) f(x_i + uh) du &= \int_{\mathbb{R}} K(u) \left(\sum_{p=0}^{P-1} \frac{h^p u^p}{p!} f^{(p)}(x_i) + \frac{h^P u^P}{P!} f^{(P)}(\tilde{x}_i) \right) du \\ &= \int_{\mathbb{R}} K(u) \left(f(x_i) + \frac{h^P u^P}{P!} f^{(P)}(\tilde{x}_i) \right) du \\ &= f(x_i) + O(h^P)\end{aligned}$$

uniformly in x_i , where $\tilde{x}_i$ is between x_i and $x_i + uh$. So by part 3,

$$\begin{aligned}\mathbb{V}[\bar{L}_n] &= \frac{4}{n} \mathbb{E}[(f(x_i) + O(h^P))^2] - \frac{4}{n} (\theta + O(h^P))^2 \\ &= \frac{4}{n} \mathbb{E}[f(x_i)^2] - \frac{4}{n} \theta^2 + O\left(\frac{h^P}{n}\right).\end{aligned}$$

Therefore $\sigma^2 = 4\mathbb{E}[f(x_i)^2] - 4\theta^2 = 4\mathbb{E}[f(x_i)^2] - 4\mathbb{E}[f(x_i)]^2$.

5. Show that $\mathbb{V}[\bar{Q}_n] = \binom{n}{2}^{-1} \frac{\mathcal{V}}{h} + o\left(\frac{1}{n^2 h}\right)$ and give the value of $\mathcal{V}$.

Solution: Recall $\mathbb{E}[Q_n(x_i, x_j) \mid x_i] = 0$, so $\mathbb{E}[Q_n(x_i, x_j) Q_n(x_r, x_l)] = 0$ unless $(i, j) = (r, l)$. Note that $\mathbb{E}[L_n(x_i)^2] = O(1)$ by part 4, hence

$$\mathbb{V}[\bar{Q}_n] = \binom{n}{2}^{-1} \mathbb{E}[Q_n(x_i, x_j)^2] = \binom{n}{2}^{-1} \mathbb{E}[U_n(x_i, x_j)^2] + O(n^{-2}).$$

Using the change of variables formula and Taylor series expansion,

$$\begin{aligned}\binom{n}{2}^{-1} \mathbb{E}[U_n(x_i, x_j)^2] &= \binom{n}{2}^{-1} h^{-1} \int_{\mathbb{R}} \int_{\mathbb{R}} K(u)^2 f(x_i + uh) du f(x_i) dx_i \\ &= \binom{n}{2}^{-1} h^{-1} \int_{\mathbb{R}} K(u)^2 du \int_{\mathbb{R}} f(x_i)^2 dx_i \\ &\quad + \binom{n}{2}^{-1} \int_{\mathbb{R}} \int_{\mathbb{R}} K(u)^2 u f^{(1)}(\tilde{x}_i) du f(x_i) dx_i \\ &= \binom{n}{2}^{-1} h^{-1} \left(\int_{\mathbb{R}} K(u)^2 du \right) \theta + O(n^{-2}).\end{aligned}$$

Therefore $\mathcal{V} = \theta \int_{\mathbb{R}} K(u)^2 du$.

6. Show that

$$\mathbb{E}[(\hat{\theta}_n - \theta)^2] = \frac{\sigma^2}{n} + \binom{n}{2}^{-1} \frac{\mathcal{V}}{h} + h^{2P} \mathcal{B}^2 + O\left(\frac{h^P}{n}\right) + o\left(\frac{1}{n^2 h} + h^{2P}\right).$$

- (i) Provide sufficient conditions on h so that $\sqrt{n}(\hat{\theta}_n - \theta) = O_{\mathbb{P}}(1)$.
- (ii) Provide sufficient conditions on h so that $\sqrt{n^2 h}(\hat{\theta}_n - \theta) = O_{\mathbb{P}}(1)$.

Solution: The result follows immediately from previous calculations, since

$$\mathbb{E}[(\hat{\theta}_n - \theta)^2] = (\mathbb{E}[\hat{\theta}_n] - \theta)^2 + \mathbb{V}[\hat{\theta}_n] = (\mathbb{E}[\hat{\theta}_n] - \theta)^2 + \mathbb{V}[\bar{L}_n] + \mathbb{V}[\bar{Q}_n].$$

Now this implies by Markov's inequality that

$$\hat{\theta}_n - \theta = O_{\mathbb{P}}\left(\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{n^2 h}} + h^P\right).$$

- (i) This holds if $\frac{1}{\sqrt{n^2 h}} \lesssim \frac{1}{\sqrt{n}}$ and $h^P \lesssim \frac{1}{\sqrt{n}}$, or equivalently if $\frac{1}{n} \lesssim h \lesssim n^{-\frac{P}{2}}$.
- (ii) This holds if $\frac{1}{\sqrt{n}} \lesssim \frac{1}{\sqrt{n^2 h}}$ and $h^P \lesssim \frac{1}{\sqrt{n^2 h}}$, or equivalently if $h \lesssim \frac{1}{n}$.

7. Give conditions on h so that $\sqrt{n}(\hat{\theta} - \theta) \rightsquigarrow \mathcal{N}(0, V_{\theta})$ and find the asymptotic variance V_{θ} .

Solution: Note that

$$\sqrt{n}(\hat{\theta}_n - \theta) = \sqrt{n}(\theta_n - \theta) + \sqrt{n}\bar{L}_n + \sqrt{n}\bar{Q}_n.$$

Using the results obtained above and Markov's inequality,

$$\begin{aligned}\sqrt{n}(\theta_n - \theta) &= O(\sqrt{n}h^P), \\ \sqrt{n}\bar{L}_n &= \frac{1}{\sqrt{n}} \sum_{i=1}^n 2(\mathbb{E}[U_n(x_i, x_j) \mid x_i] - \theta_n), \\ \sqrt{n}\bar{Q}_n &= O_{\mathbb{P}}(\sqrt{n}\sqrt{1/n^2 h}) = O_{\mathbb{P}}(1/\sqrt{nh}).\end{aligned}$$

By the central limit theorem, $\sqrt{n}\bar{L}_n \rightsquigarrow \mathcal{N}(0, V_{\theta})$ where $V_{\theta} = \sigma^2 = 4\mathbb{E}[f(x_i)^2] - 4\theta^2$. Thus the result holds whenever the other terms are negligible in probability, i.e., $\sqrt{n}h^P \rightarrow 0$ and $1/\sqrt{nh} \rightarrow 0$. Equivalently, $1/n \ll h \ll n^{-\frac{1}{2P}}$.

8. Find the mean squared error optimal bandwidth for the average density estimator $\hat{\theta}_n$, denoting it by h_{AD} . How might you feasibly estimate this in practice?

Solution: The MSE-optimal bandwidth selector is defined as $h_{\text{AD}} = \arg \min_h \mathbb{E}[(\hat{\theta}_n - \theta)^2]$. Plugging in for the expression derived in the previous part and solving the first order condition,

$$h_{\text{AD}} \in \arg \min_{h>0} \left\{ \binom{n}{2}^{-1} \frac{\mathcal{V}}{h} + h^{2P} \mathcal{B}^2 \right\} = \left(\frac{\mathcal{V}}{P\mathcal{B}^2 n(n-1)} \right)^{\frac{1}{2P+1}} \propto n^{-\frac{2}{2P+1}}.$$

To implement this procedure it is necessary to estimate $\mathcal{B}$ and $\mathcal{V}$, which could be done using semiparametric kernel-based estimators and a preliminary (or “pilot”) bandwidth.

9. Now consider the integrated mean squared error of estimating $f(x)$ using $\hat{f}_{i,n}(x)$. Assuming

for simplicity that f is compactly supported, show that

$$\int_{\mathbb{R}} \mathbb{E} \left[(\hat{f}_{i,n}(x) - f(x))^2 \right] dx = h^{2P} B^2 + \frac{V}{(n-1)h} + o\left(\frac{1}{nh} + h^{2P}\right)$$

and identify the bias B and variance V . Show that the integrated mean squared error optimal bandwidth for the nonparametric first-step, leave-one-out kernel density estimator $\hat{f}_{i,n}(x)$ is

$$h_{\text{KD}} = \left(\frac{V}{2PB^2(n-1)} \right)^{\frac{1}{2P+1}}.$$

How would you implement this in practice?

Solution: For the bias term, by a change of variables and using Taylor's theorem up to order P ,

$$\begin{aligned} \mathbb{E}[\hat{f}_{i,n}(x)] &= \int_{\mathbb{R}} \frac{1}{h} K\left(\frac{x_j - x}{h}\right) f(x_j) dx_j = \int_{\mathbb{R}} K(u) f(x + uh) du \\ &= f(x) + \frac{h^P f^{(P)}(x)}{P!} \int_{\mathbb{R}} u^P K(u) du + \frac{h^{P+1}}{(P+1)!} \int_{\mathbb{R}} f^{(P+1)}(\tilde{x}) u^{P+1} K(u) du \end{aligned}$$

where $\tilde{x}$ is between x and $x + uh$. Thus

$$\begin{aligned} &\int_{\mathbb{R}} \left(\mathbb{E}[\hat{f}_{i,n}(x)] - f(x) \right)^2 dx \\ &= \int_{\mathbb{R}} \left(\frac{h^P f^{(P)}(x)}{P!} \int_{\mathbb{R}} u^P K(u) du + \frac{h^{P+1}}{(P+1)!} \int_{\mathbb{R}} f^{(P+1)}(\tilde{x}) u^{P+1} K(u) du \right)^2 dx \\ &= \frac{h^{2P}}{(P!)^2} \int_{\mathbb{R}} f^{(P)}(x)^2 dx \left(\int_{\mathbb{R}} u^P K(u) du \right)^2 + o(h^{2P}) \end{aligned}$$

and $B^2 = \frac{1}{(P!)^2} \int_{\mathbb{R}} f^{(P)}(x)^2 dx \left(\int_{\mathbb{R}} u^P K(u) du \right)^2$ since we assume that K and $f^{(P+1)}$ are bounded and that f is compactly supported. For the variance term,

$$\begin{aligned} \mathbb{V}[\hat{f}_{i,n}(x)] &= \frac{1}{n-1} \mathbb{V} \left[\frac{1}{h} K\left(\frac{x_j - x}{h}\right) \right] \\ &= \frac{1}{n-1} \mathbb{E} \left[\frac{1}{h^2} K\left(\frac{x_j - x}{h}\right)^2 \right] - \frac{1}{n-1} \mathbb{E} \left[\frac{1}{h} K\left(\frac{x_j - x}{h}\right) \right]^2 \\ &= \frac{1}{n-1} \mathbb{E} \left[\frac{1}{h^2} K\left(\frac{x_j - x}{h}\right)^2 \right] - \frac{1}{n-1} \left(\int_{\mathbb{R}} K(u) f(x + uh) du \right)^2. \end{aligned}$$

Then by similar calculations to part 5, we see that

$$\mathbb{E} \left[\frac{1}{h^2} K\left(\frac{x_j - x}{h}\right)^2 \right] = \frac{1}{h} \int_{\mathbb{R}} K(u)^2 f(x + hu) du$$

$$= \frac{1}{h} f(x) \int_{\mathbb{R}} K(u)^2 du + \int_{\mathbb{R}} u K(u)^2 f^{(1)}(\tilde{x}) du.$$

Thus integrating gives

$$\begin{aligned} \int_{\mathbb{R}} \mathbb{V}[\hat{f}_{i,n}(x)] dx &= \frac{1}{(n-1)h} \int_{\mathbb{R}} K(u)^2 du + \frac{1}{n-1} \int_{\mathbb{R}} \int_{\mathbb{R}} u K(u)^2 f^{(1)}(\tilde{x}) dx du \\ &\quad - \frac{1}{(n-1)h} \int_{\mathbb{R}} \left(\int_{\mathbb{R}} K(u) f(x+uh) du \right)^2 dx \\ &= \frac{1}{(n-1)h} \int_{\mathbb{R}} K(u)^2 du + O(1/n) \\ &= \frac{V}{(n-1)h} + o\left(\frac{1}{nh}\right) \end{aligned}$$

with $V = \int_{\mathbb{R}} K(u)^2 du$ since $\int_{\mathbb{R}} f(x) dx = 1$ and $f^{(1)}(x)$ and K are bounded and f is compactly supported. Putting together these results,

$$\begin{aligned} \int_{\mathbb{R}} \mathbb{E} \left[(\hat{f}_{i,n}(x) - f(x))^2 \right] dx &= \int_{\mathbb{R}} (\mathbb{E}[\hat{f}_{i,n}(x)] - f(x))^2 dx + \int_{\mathbb{R}} \mathbb{V}[\hat{f}_{i,n}(x)] dx \\ &= h^{2P} B^2 + \frac{V}{(n-1)h} + o\left(\frac{1}{nh} + h^{2P}\right). \end{aligned}$$

Finally, minimizing this integrated MSE with respect to h gives the optimal bandwidth

$$h_{\text{KD}} = \left(\frac{V}{2PB^2(n-1)} \right)^{\frac{1}{2P+1}} \propto n^{-\frac{1}{2P+1}}.$$

The implementation would also require the preliminary estimation of B and V .

10. Which of the bandwidth selectors h_{AD} and h_{KD} is asymptotically larger? For each of h_{AD} and h_{KD} , explain whether the asymptotic normality result in part 7 holds.

Solution: Notice that h_{AD} is of order $n^{-2/(2P+1)}$ while h_{KD} is of order $n^{-1/(2P+1)}$. Thus $h_{\text{AD}} \ll h_{\text{KD}}$. In order for asymptotic normality to hold, we require $1/n \ll h \ll n^{-\frac{1}{2P}}$. For h_{AD} , note that $-1 < -\frac{2}{2P+1} < -\frac{1}{2P}$ so $1/n \ll h_{\text{AD}} \ll n^{-\frac{1}{2P}}$ and so asymptotic normality holds. For h_{KD} , note that $-\frac{1}{2P} < -\frac{1}{2P+1}$ so $n^{-\frac{1}{2P}} \ll h_{\text{KD}}$ and the result from part 7 fails: there is no guarantee of asymptotic normality.

This suggests that in practice we should use the undersmoothed bandwidth h_{AD} for performing inference on θ .

Question 2: Optimal Conditional Z-estimation (50 Points)

This question discusses optimal Z-estimation and inference using conditional estimating equations (i.e., moment conditions or score restrictions). Let $(\mathbf{z}_i, \mathbf{x}_i)$ for $1 \leq i \leq n$ be i.i.d. taking values in $\mathbb{R}^{d_z} \times \mathbb{R}^{d_x}$. Assume that

$$\mathbb{E}_{\boldsymbol{\theta}_0}[m(\mathbf{z}_i, \boldsymbol{\theta}) \mid \mathbf{x}_i] = 0 \text{ a.s.} \iff \boldsymbol{\theta} = \boldsymbol{\theta}_0,$$

where $\boldsymbol{\theta} \in \boldsymbol{\Theta} \subseteq \mathbb{R}^d$ and $m : \mathbb{R}^{d_z} \times \mathbb{R}^{d_x} \rightarrow \mathbb{R}$ satisfies $\mathbb{E}[m(\mathbf{z}_i, \boldsymbol{\theta})^2] < \infty$ for all $\boldsymbol{\theta} \in \boldsymbol{\Theta}$.

1. Let $\mathcal{G}$ be the class of measurable functions $\mathbf{g}$ from $\mathbb{R}^{d_x}$ to $\mathbb{R}^d$ which satisfy $\mathbb{E}[\|\mathbf{g}(\mathbf{x}_i)\|^2] < \infty$ where $\|\cdot\|$ denotes the Euclidean norm. Show that

$$\mathbb{E}_{\boldsymbol{\theta}_0}[\mathbf{g}(\mathbf{x}_i)m(\mathbf{z}_i, \boldsymbol{\theta})] = \mathbf{0} \text{ for all } \mathbf{g} \in \mathcal{G} \iff \boldsymbol{\theta} = \boldsymbol{\theta}_0.$$

Solution: For the forward direction, set $\mathbf{g}(\mathbf{x}_i) = \mathbb{E}_{\boldsymbol{\theta}_0}[m(\mathbf{z}_i, \boldsymbol{\theta}) \mid \mathbf{x}_i] \cdot \mathbf{1}_d$ to see that

$$\begin{aligned} \mathbb{E}_{\boldsymbol{\theta}_0}[\mathbf{g}(\mathbf{x}_i)m(\mathbf{z}_i, \boldsymbol{\theta})] = \mathbf{0} &\implies \mathbb{E}_{\boldsymbol{\theta}_0}[\mathbf{g}(\mathbf{x}_i)\mathbb{E}_{\boldsymbol{\theta}_0}[m(\mathbf{z}_i, \boldsymbol{\theta}) \mid \mathbf{x}_i]] = \mathbf{0} \\ &\implies \mathbb{E}_{\boldsymbol{\theta}_0}\left[\mathbb{E}_{\boldsymbol{\theta}_0}[m(\mathbf{z}_i, \boldsymbol{\theta}) \mid \mathbf{x}_i]^2 \mathbf{1}_d\right] = \mathbf{0} \\ &\implies \mathbb{E}_{\boldsymbol{\theta}_0}[m(\mathbf{z}_i, \boldsymbol{\theta}) \mid \mathbf{x}_i] = 0 \text{ a.s.} \\ &\implies \boldsymbol{\theta} = \boldsymbol{\theta}_0. \end{aligned}$$

For the backwards direction, take any $\mathbf{g} \in \mathcal{G}$ and note that setting $\boldsymbol{\theta} = \boldsymbol{\theta}_0$ gives

$$\mathbb{E}_{\boldsymbol{\theta}_0}[\mathbf{g}(\mathbf{x}_i)m(\mathbf{z}_i, \boldsymbol{\theta}_0)] = \mathbb{E}_{\boldsymbol{\theta}_0}\left[\mathbf{g}(\mathbf{x}_i)\mathbb{E}_{\boldsymbol{\theta}_0}[m(\mathbf{z}_i, \boldsymbol{\theta}_0) \mid \mathbf{x}_i]\right] = \mathbb{E}_{\boldsymbol{\theta}_0}[\mathbf{g}(\mathbf{x}_i) \cdot 0] = \mathbf{0}.$$

Consider the family of conditional Z-estimators $\hat{\boldsymbol{\theta}}_n(\mathbf{g})$ for $\mathbf{g} \in \mathcal{G}$ satisfying

$$\frac{1}{n} \sum_{i=1}^n \mathbf{g}(\mathbf{x}_i)m(\mathbf{z}_i, \hat{\boldsymbol{\theta}}_n(\mathbf{g})) = \mathbf{0}.$$

2. Suppose that $\boldsymbol{\Theta}$ is compact and $m(\mathbf{z}_i, \boldsymbol{\theta})$ is continuous in $\boldsymbol{\theta}$, uniformly in $\mathbf{z}_i$. Assume also that $\mathbf{g}$ satisfies $\boldsymbol{\theta} \neq \boldsymbol{\theta}_0 \implies \mathbb{E}_{\boldsymbol{\theta}_0}[\mathbf{g}(\mathbf{x}_i)m(\mathbf{z}_i, \boldsymbol{\theta})] \neq \mathbf{0}$. Show that $\hat{\boldsymbol{\theta}}_n(\mathbf{g}) \rightarrow_{\mathbb{P}} \boldsymbol{\theta}_0$.

Solution:

Take $\varepsilon > 0$. Since $m(\mathbf{z}_i, \cdot)$ is continuous on a compact set, uniformly in $\mathbf{z}_i$, it is uniformly continuous in $\boldsymbol{\theta}$, so there is $\delta > 0$ with $\|\boldsymbol{\theta} - \boldsymbol{\theta}'\| \leq \delta \implies |m(\mathbf{z}_i, \boldsymbol{\theta}) - m(\mathbf{z}_i, \boldsymbol{\theta}')| \leq \varepsilon$. Let $\boldsymbol{\Theta}_\delta$ be a finite δ -cover of $(\boldsymbol{\Theta}, \|\cdot\|)$. Then by the strong law of large numbers,

$$\sup_{\boldsymbol{\theta} \in \boldsymbol{\Theta}_\delta} \left\| \frac{1}{n} \sum_{i=1}^n \mathbf{g}(\mathbf{x}_i)m(\mathbf{z}_i, \boldsymbol{\theta}) - \mathbb{E}_{\boldsymbol{\theta}_0}[\mathbf{g}(\mathbf{x}_i)m(\mathbf{z}_i, \boldsymbol{\theta})] \right\| \rightarrow 0$$

almost surely. By definition of δ and again by the strong law of large numbers we also have

$$\begin{aligned} \limsup_{n \rightarrow \infty} \sup_{\|\boldsymbol{\theta} - \boldsymbol{\theta}'\| \leq \delta} \left\| \frac{1}{n} \sum_{i=1}^n \mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta}) - \frac{1}{n} \sum_{i=1}^n \mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta}') \right\| &\leq \limsup_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \|\mathbf{g}(\mathbf{x}_i)\| \varepsilon \\ &\leq \varepsilon \mathbb{E}_{\boldsymbol{\theta}_0} [\|\mathbf{g}(\mathbf{x}_i)\|] \end{aligned}$$

almost surely. Similarly

$$\sup_{\|\boldsymbol{\theta} - \boldsymbol{\theta}'\| \leq \delta} \left\| \mathbb{E}_{\boldsymbol{\theta}_0} [\mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta})] - \mathbb{E}_{\boldsymbol{\theta}_0} [\mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta}')] \right\| \leq \varepsilon \mathbb{E}_{\boldsymbol{\theta}_0} [\|\mathbf{g}(\mathbf{x}_i)\|].$$

Therefore

$$\limsup_{n \rightarrow \infty} \sup_{\boldsymbol{\theta} \in \boldsymbol{\Theta}} \left\| \frac{1}{n} \sum_{i=1}^n \mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta}) - \mathbb{E}_{\boldsymbol{\theta}_0} [\mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta})] \right\| \leq 2\varepsilon \mathbb{E}_{\boldsymbol{\theta}_0} [\|\mathbf{g}(\mathbf{x}_i)\|]$$

almost surely. Let $\varepsilon \rightarrow 0$ to see that the empirical objective function of the Z-estimator is uniformly consistent. Also observe that for all $\delta > 0$, if $\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\| \geq \delta$ then $\boldsymbol{\theta}$ lies in the compact set $\boldsymbol{\Theta} \setminus B(\boldsymbol{\theta}_0, \delta)$ and so the positive continuous function $\boldsymbol{\theta} \mapsto \|\mathbb{E}_{\boldsymbol{\theta}_0} [\mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta})]\|$ attains a positive minimum. Hence the population objective function of the Z-estimator is well-separated. Hence by the consistency theorem for Z-estimators (Van der Vaart, 2000, Theorem 5.9) we conclude that $\hat{\boldsymbol{\theta}}_n(\mathbf{g}) \rightarrow_{\mathbb{P}} \boldsymbol{\theta}_0(\mathbf{g})$.

3. Suppose the conditions of part 2 hold and let $\boldsymbol{\theta}_0$ be in the interior of $\boldsymbol{\Theta}$. Assume further that $m(\mathbf{z}_i, \boldsymbol{\theta})$ is continuously differentiable in $\boldsymbol{\theta}$ with $\sup_{\mathbf{z}_i \in \mathbb{R}^{d_z}} \left\| \frac{\partial}{\partial \boldsymbol{\theta}} m(\mathbf{z}_i, \boldsymbol{\theta}) \right\| < \infty$ for all $\boldsymbol{\theta} \in \boldsymbol{\Theta}$. We impose these conditions for the remaining parts of this question. Show that there is a non-random influence function $\boldsymbol{\psi} : \mathbb{R}^{d_z} \times \mathbb{R}^{d_x} \times \mathbb{R}^d \times \mathcal{G} \rightarrow \mathbb{R}^d$ satisfying

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_n(\mathbf{g}) - \boldsymbol{\theta}_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \boldsymbol{\psi}(\mathbf{z}_i, \mathbf{x}_i; \boldsymbol{\theta}_0, \mathbf{g}) + o_{\mathbb{P}}(1).$$

You may assume that any relevant matrices are invertible.

Solution: By the multivariate version of Taylor's theorem,

$$\begin{aligned} \mathbf{0} &= \frac{1}{n} \sum_{i=1}^n \mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \hat{\boldsymbol{\theta}}_n(\mathbf{g})) \\ &= \frac{1}{n} \sum_{i=1}^n \mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta}_0) + \frac{1}{n} \sum_{i=1}^n \mathbf{g}(\mathbf{x}_i) \frac{\partial}{\partial \boldsymbol{\theta}^\top} m(\mathbf{z}_i, \tilde{\boldsymbol{\theta}}_n(\mathbf{g})) (\hat{\boldsymbol{\theta}}_n(\mathbf{g}) - \boldsymbol{\theta}_0) \end{aligned}$$

where $\tilde{\boldsymbol{\theta}}_n = \lambda \boldsymbol{\theta}_0 + (1 - \lambda) \hat{\boldsymbol{\theta}}_n$ for some (random) $\lambda \in [0, 1]$. Rearranging and assuming the

inverse matrix exists gives

$$\widehat{\boldsymbol{\theta}}_n(\mathbf{g}) - \boldsymbol{\theta}_0 = - \left(\frac{1}{n} \sum_{i=1}^n \mathbf{g}(\mathbf{x}_i) \frac{\partial}{\partial \boldsymbol{\theta}^\top} m(\mathbf{z}_i, \widehat{\boldsymbol{\theta}}_n(\mathbf{g})) \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n \mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta}_0) \right).$$

Since $\widehat{\boldsymbol{\theta}}_n(\mathbf{g}) \rightarrow_{\mathbb{P}} \boldsymbol{\theta}_0$, by the continuous mapping theorem and the law of large numbers the first term converges in probability, giving the asymptotic linear representation

$$\widehat{\boldsymbol{\theta}}_n(\mathbf{g}) - \boldsymbol{\theta}_0 = -\mathbb{E}_{\boldsymbol{\theta}_0} \left[\mathbf{g}(\mathbf{x}_i) \frac{\partial}{\partial \boldsymbol{\theta}^\top} m(\mathbf{z}_i, \boldsymbol{\theta}_0) \right]^{-1} \left(\frac{1}{n} \sum_{i=1}^n \mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta}_0) \right) + o_{\mathbb{P}}(1).$$

The result follows with $\boldsymbol{\psi}(\mathbf{z}_i, \mathbf{x}_i; \boldsymbol{\theta}_0, \mathbf{g}) = -\boldsymbol{\Omega}(\boldsymbol{\theta}_0, \mathbf{g})^{-1} \mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta}_0)$ where

$$\boldsymbol{\Omega}(\boldsymbol{\theta}, \mathbf{g}) = \mathbb{E}_{\boldsymbol{\theta}_0} \left[\mathbf{g}(\mathbf{x}_i) \frac{\partial}{\partial \boldsymbol{\theta}^\top} m(\mathbf{z}_i, \boldsymbol{\theta}) \right].$$

4. Show that

$$\sqrt{n}(\widehat{\boldsymbol{\theta}}_n(\mathbf{g}) - \boldsymbol{\theta}_0) \rightsquigarrow \mathcal{N}(0, \mathbf{V}(\boldsymbol{\theta}_0, \mathbf{g})),$$

where $\mathbf{V}(\boldsymbol{\theta}, \mathbf{g}) = \boldsymbol{\Omega}(\boldsymbol{\theta}, \mathbf{g})^{-1} \boldsymbol{\Sigma}(\boldsymbol{\theta}, \mathbf{g}) \boldsymbol{\Omega}(\boldsymbol{\theta}, \mathbf{g})^{-\top}$ with

$$\boldsymbol{\Sigma}(\boldsymbol{\theta}, \mathbf{g}) = \mathbb{V}_{\boldsymbol{\theta}_0}[\mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta})] \quad \text{and} \quad \boldsymbol{\Omega}(\boldsymbol{\theta}, \mathbf{g}) = \mathbb{E}_{\boldsymbol{\theta}_0} \left[\mathbf{g}(\mathbf{x}_i) \frac{\partial}{\partial \boldsymbol{\theta}^\top} m(\mathbf{z}_i, \boldsymbol{\theta}) \right],$$

using the notation $\mathbf{A}^{-\top} = (\mathbf{A}^{-1})^\top$.

Solution: Using the asymptotic linear representation, note that by the central limit theorem,

$$\sqrt{n}(\widehat{\boldsymbol{\theta}}_n(\mathbf{g}) - \boldsymbol{\theta}_0(\mathbf{g})) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \boldsymbol{\psi}(\mathbf{z}_i, \mathbf{x}_i; \boldsymbol{\theta}_0, \mathbf{g}) + o_{\mathbb{P}}(1) \rightsquigarrow \mathcal{N}(0, \mathbb{V}_{\boldsymbol{\theta}_0}[\boldsymbol{\psi}(\mathbf{z}_i, \mathbf{x}_i; \boldsymbol{\theta}_0, \mathbf{g})]).$$

Finally,

$$\mathbb{V}_{\boldsymbol{\theta}_0}[\boldsymbol{\psi}(\mathbf{z}_i, \mathbf{x}_i; \boldsymbol{\theta}_0, \mathbf{g})] = \boldsymbol{\Omega}(\boldsymbol{\theta}_0, \mathbf{g})^{-1} \mathbb{V}_{\boldsymbol{\theta}_0}[\mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta}_0)] \boldsymbol{\Omega}(\boldsymbol{\theta}_0, \mathbf{g})^{-\top}.$$

5. Propose an estimator of $\mathbf{V}(\boldsymbol{\theta}_0, \mathbf{g})$, denoted by $\widehat{\mathbf{V}}_n(\mathbf{g})$, and prove it is consistent.

Solution: Let $\widehat{\mathbf{V}}_n(\mathbf{g}) = \widehat{\boldsymbol{\Omega}}_n(\widehat{\boldsymbol{\theta}}_n(\mathbf{g}), \mathbf{g})^{-1} \widehat{\boldsymbol{\Sigma}}_n(\widehat{\boldsymbol{\theta}}_n(\mathbf{g}), \mathbf{g}) \widehat{\boldsymbol{\Omega}}_n(\widehat{\boldsymbol{\theta}}_n(\mathbf{g}), \mathbf{g})^{-\top}$ where

$$\begin{aligned} \widehat{\boldsymbol{\Sigma}}_n(\boldsymbol{\theta}, \mathbf{g}) &= \frac{1}{n} \sum_{i=1}^n \left(\mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta}) \right) \left(\mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta}) \right)^\top, \\ \widehat{\boldsymbol{\Omega}}_n(\boldsymbol{\theta}, \mathbf{g}) &= \frac{1}{n} \sum_{i=1}^n \mathbf{g}(\mathbf{x}_i) \frac{\partial}{\partial \boldsymbol{\theta}^\top} m(\mathbf{z}_i, \boldsymbol{\theta}). \end{aligned}$$

To see consistency of these estimators, we argue as in the solution to part 2. Since $\widehat{\Omega}_n(\boldsymbol{\theta}, \mathbf{g})$, $\Omega(\boldsymbol{\theta}, \mathbf{g})$, $\widehat{\Sigma}_n(\boldsymbol{\theta}, \mathbf{g})$ and $\Sigma(\boldsymbol{\theta}, \mathbf{g})$ are continuous in $\boldsymbol{\theta}$ on a compact set, uniformly in $\mathbf{z}_i$, they form Glivenko–Cantelli classes. Hence we have that $\sup_{\boldsymbol{\theta} \in \Theta} |\widehat{\Sigma}_n(\boldsymbol{\theta}, \mathbf{g}) - \Sigma(\boldsymbol{\theta}, \mathbf{g})| \rightarrow 0$ a.s. and $\sup_{\boldsymbol{\theta} \in \Theta} |\widehat{\Omega}_n(\boldsymbol{\theta}, \mathbf{g}) - \Omega(\boldsymbol{\theta}, \mathbf{g})| \rightarrow 0$ a.s. By the continuous mapping theorem, we deduce that $|\Sigma(\widehat{\boldsymbol{\theta}}_n(\mathbf{g}), \mathbf{g}) - \Sigma(\boldsymbol{\theta}, \mathbf{g})| \rightarrow_{\mathbb{P}} 0$ and $|\Omega(\widehat{\boldsymbol{\theta}}_n(\mathbf{g}), \mathbf{g}) - \Omega(\boldsymbol{\theta}, \mathbf{g})| \rightarrow_{\mathbb{P}} 0$.

6. Let $\mathbf{h} : \mathbb{R}^d \rightarrow \mathbb{R}^{d_h}$ be differentiable with $\frac{\partial \mathbf{h}(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}}$ non-singular and assume $\mathbf{V}(\boldsymbol{\theta}_0, \mathbf{g})$ is invertible. Construct an asymptotically valid Wald-based hypothesis test for $H_0 : \mathbf{h}(\boldsymbol{\theta}_0) = \mathbf{0}$ against $H_1 : \mathbf{h}(\boldsymbol{\theta}_0) \neq \mathbf{0}$ at level $\alpha \in (0, 1)$.

Solution: The delta method gives

$$\sqrt{n}(\mathbf{h}(\widehat{\boldsymbol{\theta}}_n) - \mathbf{h}(\boldsymbol{\theta}_0)) \rightsquigarrow \mathcal{N}\left(0, \frac{\partial \mathbf{h}(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}^\top} \mathbf{V}(\boldsymbol{\theta}_0, \mathbf{g}) \frac{\partial \mathbf{h}(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}}\right),$$

giving the Wald test statistic

$$T_W = n(\mathbf{h}(\widehat{\boldsymbol{\theta}}_n) - \mathbf{h}(\boldsymbol{\theta}_0))^\top \left(\frac{\partial \mathbf{h}(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}^\top} \mathbf{V}(\boldsymbol{\theta}_0, \mathbf{g}) \frac{\partial \mathbf{h}(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}} \right)^{-1} (\mathbf{h}(\widehat{\boldsymbol{\theta}}_n) - \mathbf{h}(\boldsymbol{\theta}_0)) \rightsquigarrow \chi_{d_h}^2,$$

where we used the rank condition and invertibility of $\mathbf{V}(\boldsymbol{\theta}_0, \mathbf{g})$. The test rejects on the event $\{T_W > q(\chi_{d_h}^2, 1 - \alpha)\}$ where $q(\chi_{d_h}^2, \cdot)$ is the quantile function (inverse CDF) of $\chi_{d_h}^2$.

7. Let $\mathbf{c} \in \mathbb{R}^d$ be a constant vector. Construct an asymptotically valid, heteroskedasticity-robust two-sided confidence interval for $\beta_0 = \mathbf{c}^\top \boldsymbol{\theta}_0$.

Solution: By the solution to part 6, a level $1 - \alpha$ confidence interval takes the form

$$\text{CI}(\beta_0) = \left[\mathbf{c}^\top \widehat{\boldsymbol{\theta}}_n \pm z_{1-\alpha/2} \sqrt{\mathbf{c}^\top \widehat{\mathbf{V}}_n(\mathbf{g}) \mathbf{c} / n} \right]$$

where $\alpha \mapsto z_\alpha$ is the quantile function of the standard normal distribution. This standard-error estimator is heteroskedasticity-robust as we made no homoskedasticity assumptions.

8. Show that for all $\mathbf{g} \in \mathcal{G}$,

$$\mathbf{V}(\boldsymbol{\theta}_0, \mathbf{g}) - \mathbf{V}(\boldsymbol{\theta}_0, \mathbf{g}^*) = \Omega(\boldsymbol{\theta}_0, \mathbf{g})^{-1} \mathbb{E}_{\boldsymbol{\theta}_0} [\mathbf{U} \mathbf{U}^\top] \Omega(\boldsymbol{\theta}_0, \mathbf{g})^{-\top},$$

where

$$\mathbf{g}^*(\mathbf{x}_i; \boldsymbol{\theta}_0) = \frac{\mathbf{D}(\mathbf{x}_i; \boldsymbol{\theta}_0)}{\sigma^2(\mathbf{x}_i; \boldsymbol{\theta}_0)}, \quad \sigma^2(\mathbf{x}_i; \boldsymbol{\theta}) = \mathbb{V}_{\boldsymbol{\theta}_0}[m(\mathbf{z}_i, \boldsymbol{\theta}) \mid \mathbf{x}_i], \quad \mathbf{D}(\mathbf{x}_i; \boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{\theta}_0} \left[\frac{\partial}{\partial \boldsymbol{\theta}} m(\mathbf{z}_i, \boldsymbol{\theta}) \mid \mathbf{x}_i \right],$$

and

$$\mathbf{U} = \boldsymbol{\nu}_i - \Omega(\boldsymbol{\theta}_0, \mathbf{g}) \mathbb{E}_{\boldsymbol{\theta}_0} [\boldsymbol{\nu}_i^* \boldsymbol{\nu}_i^{*\top}]^{-1} \boldsymbol{\nu}_i^*, \quad \boldsymbol{\nu}_i = \mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta}_0), \quad \boldsymbol{\nu}_i^* = \mathbf{g}^*(\mathbf{x}_i; \boldsymbol{\theta}_0) m(\mathbf{z}_i, \boldsymbol{\theta}_0).$$

Solution: We drop the dependence on θ_0 for brevity of notation, writing $\mathbf{V}(\mathbf{g})$ for $\mathbf{V}(\theta_0, \mathbf{g})$, etc. Start by expanding the right-hand side in $\mathbf{U}$.

$$\begin{aligned}
& \Omega(\mathbf{g})^{-1} \mathbb{E}_{\theta_0}[\mathbf{U}\mathbf{U}^\top] \Omega(\mathbf{g})^{-\top} \\
&= \Omega(\mathbf{g})^{-1} \left(\mathbb{E}_{\theta_0}[\boldsymbol{\nu}_i \boldsymbol{\nu}_i^\top] \right) \Omega(\mathbf{g})^{-\top} \\
&\quad - \Omega(\mathbf{g})^{-1} \left(\mathbb{E}_{\theta_0}[\boldsymbol{\nu}_i \boldsymbol{\nu}_i^{*\top}] \mathbb{E}_{\theta_0}[\boldsymbol{\nu}_i^* \boldsymbol{\nu}_i^{*\top}]^{-1} \Omega(\mathbf{g})^\top \right) \Omega(\mathbf{g})^{-\top} \\
&\quad - \Omega(\mathbf{g})^{-1} \left(\Omega(\mathbf{g}) \mathbb{E}_{\theta_0}[\boldsymbol{\nu}_i^* \boldsymbol{\nu}_i^{*\top}]^{-1} \mathbb{E}_{\theta_0}[\boldsymbol{\nu}_i^* \boldsymbol{\nu}_i^\top] \right) \Omega(\mathbf{g})^{-\top} \\
&\quad + \Omega(\mathbf{g})^{-1} \left(\Omega(\mathbf{g}) \mathbb{E}_{\theta_0}[\boldsymbol{\nu}_i^* \boldsymbol{\nu}_i^{*\top}]^{-1} \mathbb{E}_{\theta_0}[\boldsymbol{\nu}_i^* \boldsymbol{\nu}_i^{*\top}] \mathbb{E}_{\theta_0}[\boldsymbol{\nu}_i^* \boldsymbol{\nu}_i^{*\top}]^{-1} \Omega(\mathbf{g})^\top \right) \Omega(\mathbf{g})^{-\top} \\
&= \mathbf{V}(\mathbf{g}) - \mathbb{E}_{\theta_0}[\boldsymbol{\nu}_i^* \boldsymbol{\nu}_i^{*\top}]^{-1} \\
&= \mathbf{V}(\mathbf{g}) - \mathbf{V}(\mathbf{g}^*)
\end{aligned}$$

because

$$\begin{aligned}
\mathbb{E}_{\theta_0}[\boldsymbol{\nu}_i \boldsymbol{\nu}_i^{*\top}] &= \mathbb{E}_{\theta_0} \left[\mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \theta_0) (\mathbf{g}^*(\mathbf{x}_i; \theta_0) m(\mathbf{z}_i, \theta_0))^\top \right] \\
&= \mathbb{E}_{\theta_0} \left[\mathbf{g}(\mathbf{x}_i) \frac{\mathbf{D}(\mathbf{x}_i; \theta_0)^\top}{\sigma^2(\mathbf{x}_i; \theta_0)} m(\mathbf{z}_i, \theta_0)^2 \right] \\
&= \mathbb{E}_{\theta_0} \left[\mathbf{g}(\mathbf{x}_i) \frac{\mathbf{D}(\mathbf{x}_i; \theta_0)^\top}{\sigma^2(\mathbf{x}_i; \theta_0)} \mathbb{E}_{\theta_0} \left[m(\mathbf{z}_i, \theta_0)^2 \mid \mathbf{x}_i \right] \right] = \Omega(\mathbf{g})
\end{aligned}$$

since

$$\begin{aligned}
\Sigma(\mathbf{g}) &= \mathbb{V}_{\theta_0}[\mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \theta_0)] = \mathbb{E}_{\theta_0} \left[\mathbb{E}_{\theta_0} [m(\mathbf{z}_i, \theta_0)^2 \mid \mathbf{x}_i] \mathbf{g}(\mathbf{x}_i) \mathbf{g}(\mathbf{x}_i)^\top \right] \\
&= \mathbb{E}_{\theta_0} [\sigma^2(\mathbf{x}_i; \theta_0) \mathbf{g}(\mathbf{x}_i) \mathbf{g}(\mathbf{x}_i)^\top]
\end{aligned}$$

and

$$\begin{aligned}
\mathbb{E}_{\theta_0}[\boldsymbol{\nu}_i^* \boldsymbol{\nu}_i^{*\top}] &= \mathbb{E}_{\theta_0} \left[\mathbb{E}_{\theta_0} [m(\mathbf{z}_i, \theta_0)^2 \mid \mathbf{x}_i] \mathbf{g}^*(\mathbf{x}_i; \theta_0) \mathbf{g}^*(\mathbf{x}_i; \theta_0)^\top \right] \\
&= \mathbb{E}_{\theta_0} \left[\sigma^2(\mathbf{x}_i; \theta_0) \frac{\mathbf{D}(\mathbf{x}_i; \theta_0) \mathbf{D}(\mathbf{x}_i; \theta_0)^\top}{\sigma^4(\mathbf{x}_i; \theta_0)} \right] \\
&= \mathbb{E}_{\theta_0} \left[\frac{\mathbf{D}(\mathbf{x}_i; \theta_0) \mathbf{D}(\mathbf{x}_i; \theta_0)^\top}{\sigma^2(\mathbf{x}_i; \theta_0)} \right]
\end{aligned}$$

and

$$\begin{aligned}
\mathbf{V}(\mathbf{g}^*) &= \Omega(\mathbf{g}^*)^{-1} \Sigma(\mathbf{g}^*) \Omega(\mathbf{g}^*)^{-1} \\
&= \mathbb{E}_{\theta_0} \left[\frac{\mathbf{D}(\mathbf{x}_i; \theta_0) \mathbf{D}(\mathbf{x}_i; \theta_0)^\top}{\sigma^2(\mathbf{x}_i; \theta_0)} \right]^{-1} \mathbb{E}_{\theta_0} \left[\frac{\mathbf{D}(\mathbf{x}_i; \theta_0) \mathbf{D}(\mathbf{x}_i; \theta_0)^\top}{\sigma^2(\mathbf{x}_i; \theta_0)} \right] \mathbb{E}_{\theta_0} \left[\frac{\mathbf{D}(\mathbf{x}_i; \theta_0) \mathbf{D}(\mathbf{x}_i; \theta_0)^\top}{\sigma^2(\mathbf{x}_i; \theta_0)} \right]^{-1} \\
&= \mathbb{E}_{\theta_0} \left[\frac{\mathbf{D}(\mathbf{x}_i; \theta_0) \mathbf{D}(\mathbf{x}_i; \theta_0)^\top}{\sigma^2(\mathbf{x}_i; \theta_0)} \right]
\end{aligned}$$

since

$$\begin{aligned}\Omega(\mathbf{g}) &= \mathbb{E}_{\theta_0} \left[\mathbf{g}(\mathbf{x}_i) \frac{\partial}{\partial \boldsymbol{\theta}^\top} m(\mathbf{z}_i, \boldsymbol{\theta}_0) \right] = \mathbb{E}_{\theta_0} \left[\mathbf{g}(\mathbf{x}_i) \mathbb{E}_{\theta_0} \left[\frac{\partial}{\partial \boldsymbol{\theta}^\top} m(\mathbf{z}_i, \boldsymbol{\theta}_0) \mid \mathbf{x}_i \right] \right] \\ &= \mathbb{E}_{\theta_0} [\mathbf{g}(\mathbf{x}_i) \mathbf{D}(\mathbf{x}_i; \boldsymbol{\theta}_0)^\top].\end{aligned}$$

9. Show that under the semi-definite partial ordering defined by $\mathbf{V} \preceq \mathbf{V}'$ if and only if $\mathbf{V}' - \mathbf{V}$ is positive semi-definite, we have

$$\mathbf{g}^*(\mathbf{x}_i; \boldsymbol{\theta}_0) \in \arg \min_{\mathbf{g} \in \mathcal{G}} \mathbf{V}(\boldsymbol{\theta}_0, \mathbf{g}).$$

Propose a feasible, asymptotically efficient estimator of $\boldsymbol{\theta}_0$.

Solution: By the previous part, $\mathbf{V}(\boldsymbol{\theta}_0, \mathbf{g}^*) \preceq \inf_{\mathbf{g} \in \mathcal{G}} \mathbf{V}(\boldsymbol{\theta}_0, \mathbf{g})$ because $\mathbb{E}_{\theta_0}[\mathbf{U}\mathbf{U}^\top] \succeq 0$. Hence $\mathbf{g}^*$ minimizes the asymptotic variance of $\hat{\boldsymbol{\theta}}_n(\mathbf{g})$ in the class $\mathcal{G}$. To get a feasible efficient estimator $\hat{\boldsymbol{\theta}}_n$, we can solve the Z-estimation problem

$$\frac{1}{n} \sum_{i=1}^n \frac{\hat{\mathbf{D}}(\mathbf{x}_i; \hat{\boldsymbol{\theta}}_n)}{\hat{\sigma}^2(x_i; \hat{\boldsymbol{\theta}}_n)} m(\mathbf{z}_i, \hat{\boldsymbol{\theta}}_n) = \mathbf{0}$$

where $\hat{\mathbf{D}}(\mathbf{x}_i; \boldsymbol{\theta})$ and $\hat{\sigma}^2(x_i; \boldsymbol{\theta})$ would need to be estimated in a preliminary step. This approach is sometimes called “continuously updated” because the optimal weighting is estimated at the same time as the parameter of interest.

10. Suppose that $\mathbf{z}_i = (y_i, \mathbf{x}_i)$ takes values in $\mathbb{R} \times \mathbb{R}^{d_x}$ and assume that it satisfies $y_i = \mathbf{x}_i^\top \boldsymbol{\theta}_0 + \varepsilon_i$ with $\mathbb{E}_{\theta_0}[\varepsilon_i \mid \mathbf{x}_i] = 0$ and $v(\mathbf{x}_i) = \mathbb{E}[\varepsilon_i^2 \mid \mathbf{x}_i]$. Consider the least-squares estimator $\hat{\boldsymbol{\theta}}_{\text{LS}}$ and the weighted least-squares estimator $\hat{\boldsymbol{\theta}}_{\text{WLS}}$ defined by the first order conditions

$$\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i (y_i - \mathbf{x}_i^\top \hat{\boldsymbol{\theta}}_{\text{LS}}) = \mathbf{0} \quad \text{and} \quad \frac{1}{n} \sum_{i=1}^n \frac{\mathbf{x}_i}{v(\mathbf{x}_i)} (y_i - \mathbf{x}_i^\top \hat{\boldsymbol{\theta}}_{\text{WLS}}) = \mathbf{0}$$

respectively. Compute the asymptotic variances of $\hat{\boldsymbol{\theta}}_{\text{LS}}$ and $\hat{\boldsymbol{\theta}}_{\text{WLS}}$ and discuss their relative asymptotic efficiencies. *Hint: You may wish to identify the optimal Z-estimator $\hat{\boldsymbol{\theta}}_n(\mathbf{g}^*)$.*

Solution: For $\hat{\boldsymbol{\theta}}_{\text{LS}}$, note that $\mathbf{g}(\mathbf{x}_i) = \mathbf{x}_i$ and $m(\mathbf{z}_i, \boldsymbol{\theta}) = y_i - \mathbf{x}_i^\top \boldsymbol{\theta}$ so that $m(\mathbf{z}_i, \boldsymbol{\theta}_0) = \varepsilon_i$. By part 4, the asymptotic variance of this estimator is

$$\mathbf{V}(\hat{\boldsymbol{\theta}}_{\text{LS}}) = \mathbb{E} [\mathbf{x}_i \mathbf{x}_i^\top]^{-1} \mathbb{E} [\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i^\top] \mathbb{E} [\mathbf{x}_i \mathbf{x}_i^\top]^{-1}.$$

Next, we show that $\hat{\boldsymbol{\theta}}_{\text{WLS}} = \hat{\boldsymbol{\theta}}_n(\mathbf{g}^*)$ is the optimal Z-estimator. Set $m(\mathbf{z}_i, \boldsymbol{\theta}) = y_i - \mathbf{x}_i^\top \boldsymbol{\theta}$ again. By part 8, we have

$$\mathbf{D}(\mathbf{x}_i; \boldsymbol{\theta}) = \mathbb{E}_{\theta_0} \left[\frac{\partial}{\partial \boldsymbol{\theta}} m(\mathbf{z}_i, \boldsymbol{\theta}) \mid \mathbf{x}_i \right] = \mathbf{x}_i$$

and

$$\sigma^2(\mathbf{x}_i; \boldsymbol{\theta}) = \mathbb{V}_{\boldsymbol{\theta}_0}[m(\mathbf{z}_i, \boldsymbol{\theta}) \mid \mathbf{x}_i] = \mathbb{V}_{\boldsymbol{\theta}_0}[y_i \mid \mathbf{x}_i] = \mathbb{E}_{\boldsymbol{\theta}_0}[\varepsilon_i^2 \mid \mathbf{x}_i] - \mathbb{E}_{\boldsymbol{\theta}_0}[\varepsilon_i \mid \mathbf{x}_i]^2 = v(\mathbf{x}_i)$$

which gives

$$\mathbf{g}^*(\mathbf{x}_i; \boldsymbol{\theta}) = \frac{\mathbf{x}_i}{v(\mathbf{x}_i)}.$$

Then the optimal Z-estimator $\hat{\boldsymbol{\theta}}_n(\mathbf{g}^*)$ satisfies

$$0 = \frac{1}{n} \sum_{i=1}^n \mathbf{g}^*(\mathbf{x}_i) m(\mathbf{z}_i, \hat{\boldsymbol{\theta}}_n(\mathbf{g}^*)) = \frac{1}{n} \sum_{i=1}^n \frac{\mathbf{x}_i (y_i - \mathbf{x}_i^\top \hat{\boldsymbol{\theta}}_n(\mathbf{g}^*))}{v(\mathbf{x}_i)}$$

and so $\hat{\boldsymbol{\theta}}_n(\mathbf{g}^*) = \hat{\boldsymbol{\theta}}_{\text{WLS}}$. We plug into the asymptotic variance derived in part 4 to see

$$\mathbf{V}(\hat{\boldsymbol{\theta}}_{\text{WLS}}) = \mathbb{E}_{\boldsymbol{\theta}_0} \left[\frac{\mathbf{x}_i \mathbf{x}_i^\top}{v(\mathbf{x}_i)} \right]^{-1} \mathbb{E}_{\boldsymbol{\theta}_0} \left[\frac{\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i^\top}{v(\mathbf{x}_i)^2} \right] \mathbb{E}_{\boldsymbol{\theta}_0} \left[\frac{\mathbf{x}_i \mathbf{x}_i^\top}{v(\mathbf{x}_i)} \right]^{-1} = \mathbb{E}_{\boldsymbol{\theta}_0} \left[\frac{\mathbf{x}_i \mathbf{x}_i^\top}{v(\mathbf{x}_i)} \right]^{-1}.$$

Since $\hat{\boldsymbol{\theta}}_{\text{WLS}}$ uses the optimal function $\mathbf{g}^*$, by part 8 we have

$$\mathbf{V}(\hat{\boldsymbol{\theta}}_{\text{WLS}}) \preceq \mathbf{V}(\hat{\boldsymbol{\theta}}_{\text{LS}}).$$

Equality holds when $v(\mathbf{x}_i) = v$ does not depend on $\mathbf{x}_i$.

References

Van der Vaart, A. W. (2000). *Asymptotic Statistics*, volume 3. Cambridge University Press.